TOP 10 PYSPARK TRANSFORMATIONS & ACTIONS EVERY DATA ENGINEER SHOULD KNOW

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WHAT ARE TRANSFORMATIONS & ACTIONS?

Transformations

Operations that create a new DataFrame/RDD from an existing one.

- Lazy Execution
- Build a DAG (Directed Acyclic Graph)
- No computation happens immediately

Actions

Operations that trigger execution and return results.

- Start actual computation
- Execute the DAG built by transformations
- Return results or write data to storage

Common Transformations:

- select()
- filter()
- withColumn()
- join()
- groupBy()
- orderBy()

Common Actions:

- show()
- count()
- collect()
- first()
- take()
- write()

SELECT()

Extract only the columns needed for analysis.

- Reduces data movement
- Improves performance
- Makes pipelines cleaner

Example
Selecting customer ID and transaction amount from a billion-row sales dataset.

```
df.select("id", "name", "salary")
```

FILTER() / WHERE()

Keep only rows matching a condition.

Data cleaning
Data validation
Faster downstream processing

Use Case
Filtering active users before generating reports.

```
# filter
df.filter(df.salary > 50000)

# where
df.where("salary > 50000")
```

WITHCOLUMN()

Add derived columns or transform existing ones.

- Feature Engineering
- Data Enrichment
- Business Calculations

Use Case
Creating revenue, profit, tax, or KPI columns.

```
from pyspark.sql.functions import col

df.withColumn(
    "annual_salary",
    col("salary") * 12
)
```

GROUPBY() + AGG()

Summarize large datasets.

- Business Reporting
- KPI Calculation
- Dashboard Metrics

Example

- Average Salary
- Total Revenue
- Customer Count
- Product Sales

```
df.groupBy("department") \
  .agg({"salary":"avg"})
```

JOIN()

Merge data from different sources.

Join Types

- Inner Join
- Left Join
- Right Join
- Full Join

Real-World Example
Combining Orders, Customers, and Products tables.

```
orders.join(
 customers,
 "customer_id",
 "inner"
)
```

DISTINCT() & DROPDUPLICATES()

Remove duplicate records.

- Better Data Quality
- Accurate Reporting
- Reliable Analytics

Use Case
Removing duplicate customer registrations.

```
# distinct
df.distinct()

#dropDuplicates
df.dropDuplicates(["email"])
```

COUNT()

Count records in a dataset.

- Data Validation
- Record Reconciliation
- Pipeline Monitoring

Example
Verifying source and target row counts after ETL.

```
df.count()
```

SHOW(), COLLECT() & WRITE()

show() - Display rows from a DataFrame for inspection.

collect() - Retrieve all records from executors to the driver node.

write() - Persist DataFrames to storage systems. Save processed data permanently

```
# show
df.show(5)

#collect
data = df.collect()

# write
df.write.parquet("/data/output")
```

THANK YOU

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